

Program -10-Documentation

Title: Demonstrate Kubernetes Dashboard using CLI and Kubernetes Cluster

Objective: Deploy an Nginx application, expose it via a service, and monitor the infrastructure using the Kubernetes Dashboard.

1. Project Initialization

Before applying configurations, create a dedicated workspace to keep your YAML files organized.

- `mkdir new-progr10`: Creates a new directory for the project.
- `cd new-program-10`: Moves the terminal context into the project folder.

2. Configuration Creation

The nano command is used here as a text editor to define the desired state of your cluster.

`nano deployment.yaml`: Used to define the Deployment (specifying the Nginx image, number of replicas, and labels).

`nano services.yaml`: Used to define the Service (specifying how to expose the pods to external traffic using a NodePort).

3. Applying Configurations

The `kubectl apply` command sends your YAML definitions to the Kubernetes API server.

Command: `kubectl apply -f deployment.yaml`

- Purpose: Creates the Pods based on the deployment strategy. In your case, it resulted in 2 running Nginx pods.

Command: `kubectl apply -f services.yaml`

- Purpose: Creates a logical shim (Service) that routes traffic to pods labeled with `app: nginx`.

4. Verification and Status Checks

- **Command**

- kubectl get pods
- kubectl get services
- kubectl get deployments

Purpose

Lists all active pods. Shows status (e.g., Running), readiness, and age.

5. Accessing the Application

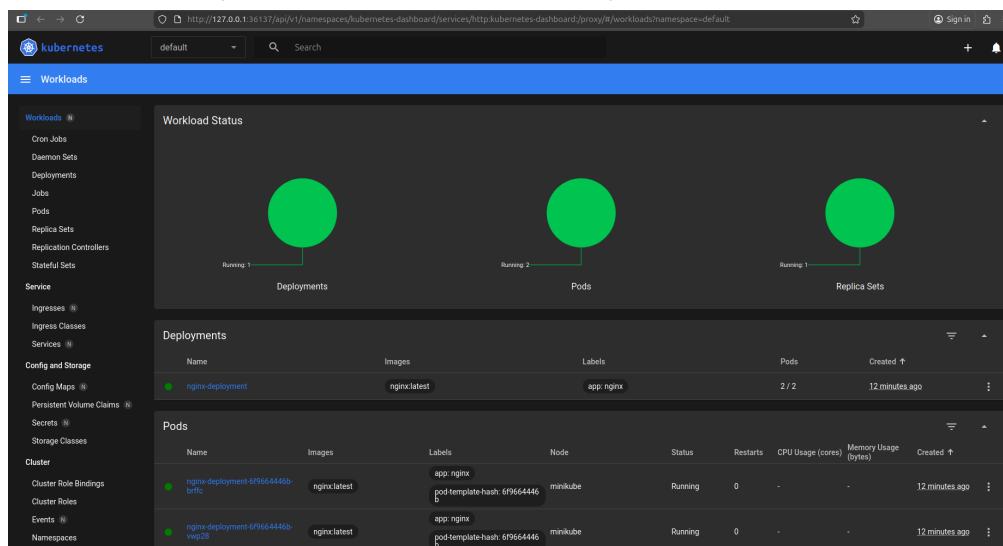
- **Command:** minikube service nginx-service

This generates a local URL (e.g., <http://192.168.49.2:32649>) and automatically opens your default web browser to the "Welcome to nginx!" page.

6. Cluster Visualization

- **Command:** minikube dashboard

It will display all the Status of deployment ,pods , replica set



1rv24mc082_rajatraj@rajatraj-Vostro-5581: ~/devops/labprogram/prgm9

1rv24mc082_rajatraj@rajatraj-Vostro-5581: ~/devops/labprogram/prgm10

1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/Labprogram/prgm105 minikube service nginx-service

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:31516

🔥 Opening service default/nginx-service in default browser...

1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/Labprogram/prgm105 Gtk-Message: 21:11:22.191: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.

[23457, Main Thread] WARNING: GTK+ module /snap/firefox/7559/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so cannot be loaded.

GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.: 'glib warning', file /build/firefox/parts/firefox/build/toolkit/xre/nsSigHandlers.cpp:201

(firefox_firefox:23457): Gtk-WARNING **: 21:11:22.265: GTK+ module /snap/firefox/7559/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so cannot be loaded.

GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.

Gtk-Message: 21:11:22.265: Failed to load module "canberra-gtk-module"

[23457, Main Thread] WARNING: GTK+ module /snap/firefox/7559/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so cannot be loaded.

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1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/Labprogram/prgm105 minikube service nginx-service

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:31516

🔥 Opening service default/nginx-service in default browser...

1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/Labprogram/prgm105 Gtk-Message: 21:16:44.885: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.

[26149, Main Thread] WARNING: GTK+ module /snap/firefox/7559/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so cannot be loaded.

GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.: 'glib warning', file /build/firefox/parts/firefox/build/toolkit/xre/nsSigHandlers.cpp:201

(firefox_firefox:26149): Gtk-WARNING **: 21:16:45.028: GTK+ module /snap/firefox/7559/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so cannot be loaded.

GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.

Gtk-Message: 21:16:45.028: Failed to load module "canberra-gtk-module"

[26149, Main Thread] WARNING: GTK+ module /snap/firefox/7559/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so cannot be loaded.

GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.: 'glib warning', file /build/firefox/parts/firefox/build/toolkit/xre/nsSigHandlers.cpp:201

(firefox_firefox:26149): Gtk-WARNING **: 21:16:45.031: GTK+ module /snap/firefox/7559/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so cannot be loaded.

GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.

Gtk-Message: 21:16:45.031: Failed to load module "canberra-gtk-module"